Thomas Russell

AEROSPACE ENGINEERING AND SCIENCE GRADUATE

+61 414 084 500
TOMMYCPRUSSELL@GMAIL.COM
VICTORIA, AUSTRALIA

SKILLS & ABILITIES

Software & CAD/CAM: Siemens NX (+ Teamcenter), SolidWorks 2025, PTC Creo (+ Windchill PLM), Fusion 360 CAM, KiCad
Simulation & DAQ: ANSYS (FEA/CFD), NX Simcenter (NASTRAN), MATLAB & Simulink, Python, C++, C#, JavaScript, NI LabVIEW, LabJack
Manufacturing: Machining (CNC/Manual), Additive Manufacturing, Welding, Composites, Drawing (ASME Y14.5)

PROFESSIONAL EXPERIENCE

Rocket Lab Development Engineer, Neutron Thrust Module *Summer 2025 – 14-week internship*

- Executed structural analysis and sub-system testing on the thrust module including Archimedes interfaces and feed systems to drive validation and flight qualification.
- Owned the complete component lifecycle, from conceptual design and CAD modeling to final drawing packages and manufacturing release. (Read more)

Monash High Powered Rocketry (MHPR) *Since 2022*

- **Team Lead** (*2025–2026*): Directing 150+ students targeting a 100,000 feet hybrid rocket launch and competing in the AURC 10k ft SRAD category; secured over $85,000 in sponsorships.
- **Propulsion Lead** (*2024–2025*): Led design, manufacture and test of 3kN (Read more) and 12kN (cryogenic liquid oxygen, Read more) hybrid engines, winning the IREC Jim Furfaro Award for Technical Excellence. Includes all ground support equipment and test infrastructure.
- Engineered composite over-wrapped pressure vessels, regeneratively cooled nozzles, and custom pneumatic oxidiser systems, taking hardware from concept to completion.

Hydrogen Jet Facility, Final Year Project *2026 – 2 semesters*

- Design, build and commission of a hydrogen jet facility at Monash University. Responsible for implementing all fluids systems, control, data acquisition and safety measures for the facility.
- Investigating supersonic hydrogen jet leaks with acoustics involving the use of high speed schlieren imaging techniques and subsequent analysis. (Read more)

Liquid Rocket Engine Project *2026 – Personal project*

- Designed and built a nitrous oxide fill system including fluids, housing, custom PCB, complete networking and DAQ.
- Design and test of a N2O/IPA engine featuring a custom pyro-actuated Urbanski-Colburn valve and pintle injector.
- Characterised valve and injector empirical discharges. Developed a complete thermodynamic simulation for the entire engine to predict performance.
- Hot fire completed, exceeding performance expectations with C* efficiency over 80%. (Read more)

Summer Research Project, Shock Lab Research Group *Summer 2024 – 10-week program*

- Designed, manufactured, and tested a planar glass rocket nozzle to investigate Free Shock Separation (FSS) phenomena.
- Executed complex CFD analysis and processed high-speed schlieren imaging data utilising Spectral Proper Orthogonal Decomposition (SPOD). (Read more)

Assistant Robot Operator, Titomic *Since 2024*

- Operating 6-axis robots equipped with cold spray additive manufacturing heads to produce commercial components.
- Conducting metallurgical R&D on novel alloys, preparing and analysing samples via high-powered microscopy to evaluate material properties.
- Machining product scaffolds, tensile coupons, and post-sprayed items using manual and CNC techniques.

QUALIFICATIONS AND AWARDS

B.E (Honours) / B.S (Astrophysics) - *5 yr (Washington Accord / ABET accredited; Master's equivalent)* 2026
Jim Furfaro Award for Technical Excellence, IREC 2025 *2025*
Runner up, 10k feet SRAD hybrid category, IREC 2025 *2025*
Machining short course, Chisholm Institute *2023*
ANSYS FEA and CFD Introduction course, Leap Australia 2023